II.1 Q1

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When $N=0$, $a \geq 1 \geq F_{0+1}$ and $b \geq 1 \geq F_{0+2}$ since $a, b \in \mathbb{Z}^+$

Assume that when $N=k$, if the algorithm takes k divisions to reach the answer, then $a \geq F_{k+2}$ and $b \geq F_{k+1}$

Say the first division of the algorithm is of the form $a = qb + r$, $0 \leq r \leq b$

Then in the $N=k+1$ case, the first step needs to output $A = Qa + b$ with $0 \leq b \leq a$, where A and a are the numbers to find the hcf of

From the assumption, $a \geq F_{k+2} \geq F_{(k+1)+1}$ as required.

$$
\begin{aligned}
Q \geq 1, \text{ so } A &\geq a + b \\
&\geq F_{k+2} + F_{k+1} \quad \text{by the assumption} \\
&\geq F_{k+3} \\
&\geq F_{(k+1)+2} \quad \text{as required}
\end{aligned}
$$

So by induction, since the relationship in question holds for $N=0$, and also in the $N=k+1$ case when $N=k$ is true, it must therefore be true $\forall\, N \in \mathbb{N}$